

CONSOLIDATED ENGINEERING AND SOURCING CONSULTANTS (CESC)

Engineering White Paper

Substation Arc-Flash Mitigation

Engineering Methods for MV and LV Systems, Incident-Energy Reduction, and Retrofit Improvement

Prepared, reviewed, and issued by CESC

Website publication edition

Executive abstract

This white paper addresses the engineering methods used to reduce arc-flash exposure in substations and industrial switchgear installations, including protection speed improvements, arc-resistant equipment strategy, remote operation, zonal interlocking, and operational controls.

It also introduces practical engineering checks used during concept design, detailed design, and retrofit evaluation so that mitigation decisions are tied to measurable reduction in incident energy, clearing time, and operator exposure.

Document basis

Document type	Technical white paper
Primary basis	IEEE 1584, NFPA 70E, IEC 62271-200, IEC 61439...
Intended audience	Utilities, EPC contractors, industrial asset owners, and engineering teams
Publication note	Prepared in formal issuer style to demonstrate CESC engineering capability in studies, protection, digital integration, retrofit, asset management, and brownfield risk-reduction engineering.

Applicable references

Reference	Application in this paper
IEEE 1584	Arc-flash incident-energy study methodology.
NFPA 70E	Electrical safety work-practice framework.
IEC 62271-200	MV switchgear context and station implementation environment.
IEC 61439	LV assembly context where LV systems are included in arc-flash review.
IEC 61936-1	Power installation engineering and general station design context.
IEEE C37.20.7	Arc-resistant switchgear concepts and accessibility context.
IEC 60255 series	Protection relay performance and trip philosophy context.

Risk context

Arc-flash mitigation is not only a safety compliance matter. It also drives design decisions for switchgear specification, relay selection, compartmentation, maintenance philosophy, and operating procedures.

In practical projects, the lowest-risk solution is usually a combination of equipment design, protective-device speed, remote working methods, and controlled maintenance access.

Typical high-risk conditions include high fault current at low system impedance, slow upstream backup clearing, withdrawn compartment shutters defeated during maintenance, and LV boards with long device clearing times. These conditions should be identified early because the mitigation route is different for each case.

Engineering hierarchy of mitigation

The mitigation hierarchy should be treated in layers: reduce the available fault energy where practical, clear the fault faster, increase personnel distance, limit exposure time, and improve enclosure behavior under internal arc conditions.

For MV switchgear, internal-arc classification under the relevant standard remains important, but it is not a substitute for protection design and safe operating practice.

In engineering terms, internal-arc classification improves enclosure behaviour, but the incident-energy study still depends strongly on system fault current and total arc duration. A robust design therefore combines enclosure performance with fast tripping philosophy and practical operating controls.

Calculation framework

Arc-flash studies generally begin with fault level, clearing time, working distance, and equipment configuration. The most influential adjustable parameter in many installations is clearing time, which is why selective protection must be balanced against fast energy reduction at close-in faults.

From an engineering perspective, the design target is to reduce incident energy by reducing let-through time while preserving dependable operation for external faults and downstream selectivity.

Practical engineering calculation example 1 - impact of clearing time: assume an MV switchboard working distance of 900 mm, arcing current derived from the study model of about 22 kA, and two operating cases. Case A clears in 0.40 s by time-graded overcurrent only. Case B clears in 0.08 s using arc-flash detection or high-speed bus protection. For the same equipment and fault level, incident energy varies approximately in proportion to arc duration, so reducing clearing time from 0.40 s to 0.08 s cuts the energy term by about 80 percent. This is why protection modernization often produces a greater risk reduction than label updates alone.

Practical engineering calculation example 2 - relay reach and selectivity check: if a feeder protection stage is intentionally delayed to preserve downstream coordination, the engineer should confirm whether a separate high-speed element can be applied only for bus or compartment faults. As a simple check, if the normal selective stage operates at 350 ms and a dedicated arc-flash element trips at 60 ms for internal faults, the busbar section sees nearly a 5.8 times reduction in fault duration. This reduction is often sufficient to move the installation into a materially safer maintenance condition without disturbing downstream discrimination for external faults.

Practical engineering calculation example 3 - fault level screening: for preliminary design, a first-pass three-phase short-circuit current can be estimated from $I = S / (\sqrt{3} \times V)$. For example, a 25 MVA source on an 11 kV bus gives approximately 1,312 A source contribution before impedance refinement, while a stronger utility source of 250 MVA gives about 13.1 kA. This simple screening step helps the engineer identify which buses require more urgent incident-energy review and whether current-limiting or faster protection concepts should be examined during early design.

Typical mitigation measures

Common engineered measures include arc-flash detection combined with high-speed trip, bus differential protection, zone-selective interlocking, maintenance mode settings, remote racking, remote switching, and use of arc-resistant switchgear where justified.

For retrofit projects, one of the highest-value actions is often protection modernization because it can materially shorten clearing time without full switchgear replacement.

Practical engineering application example 4 - zone-selective interlocking: if an LV incomer is set for 500 ms to maintain selectivity and a downstream feeder detects the fault and restrains the incomer, the system keeps coordination for feeder faults. If the fault occurs on the main bus and no restraining signal is received, the incomer can trip almost instantaneously. In practice, this approach preserves system discrimination while sharply reducing incident energy on the bus section where personnel exposure is highest.

Practical engineering application example 5 - remote operation value: if the calculated incident-energy boundary extends beyond the normal operating position, remote switching or remote racking changes the risk picture even when the fault energy itself is unchanged. In engineering terms, increased operator distance and removal of the person from the arc zone may be the most economical brownfield mitigation where replacing switchgear is not justified.

Execution considerations

A mitigation program should be documented through single-line review, protection setting review, operating scenario review, labeling policy, and maintenance procedure update. Clients should avoid treating labels as the final output; labels are only the visible end of a wider engineering assessment.

Constructability also matters. If a mitigation concept depends on complex bypass states or poorly understood settings groups, the operational benefit may be lost in practice.

A sound project deliverable should therefore include operating scenario matrices, settings philosophy, interlock review, maintenance-mode entry and exit logic, and testing steps for each mitigation function. Without these execution documents, even a technically correct concept can fail to deliver the expected field safety improvement.

Typical verification checks include confirmation of CT polarity for differential or arc-detection logic, breaker trip coil health, total clearing time from relay pickup to contact parting, and proof that any maintenance mode cannot be left enabled unintentionally. These are practical engineering checks that determine whether theoretical study gains will actually be achieved on site.

CESC engineering contribution

CESC can support arc-flash risk reduction through system studies, equipment suitability review, protection upgrade scope definition, operational scenario analysis, and practical retrofit strategies that improve safety without unnecessary plant replacement.

CESC value addition is strongest where clients need an engineering decision rather than a software printout. This includes comparing alternative mitigation paths, checking whether a protection upgrade can achieve the target reduction before switchgear replacement is considered, and defining brownfield implementation steps that remain workable during outages, commissioning, and future maintenance.

CESC can also translate study results into implementable scope by aligning incident-energy findings with relay settings, interlocking philosophy, equipment limitations, outage sequencing, FAT and SAT requirements, and asset-life considerations. This practical integration is where project value is usually gained or lost.

Conclusion

These website papers are structured to demonstrate how CESC approaches engineering subjects at system level, not only at equipment catalogue level. In arc-flash mitigation work, the real engineering value lies in converting study outputs into a workable plant solution: selecting the right protection architecture, checking the achievable clearing-time improvement, confirming field constructability, and balancing safety gain against brownfield cost and outage constraint. CESC adds value by connecting calculations, protection design, retrofit scope, testing philosophy, and operational practicality into one coherent engineering package that clients can confidently implement.